Abhinav S

Chennai, Tamil Nadu | vanihba.sv@gmail.com | 9551892619 | linkedin | github.com/Thundersnow18

Summary

Computer Science undergraduate with solid knowledge in data structures, algorithms, and system design. Skilled in full-stack development using contemporary technologies such as Next.js, Node.js, and PostgreSQL, having real-world exposure to designing scalable backend systems. Experienced in tackling challenging problems in web development as well as machine learning tasks.

Education

B.Tech. Computer Science and Engineering (Cyber Security) — Aug 2023 – July 2027
SRM Institute of Science and Technology — *Chennai, Tamil Nadu*
- CGPA: 9.85

Class 12 (CBSE) — 2022 – 2023
Sri Chaitanya — *Chennai, Tamil Nadu*
- Scored 89%

Class 10 (CBSE) — 2020 – 2021
Narayana E Techno — *Chennai, Tamil Nadu*
- Scored 88%

Projects

Giftwise | *Next.js, Node.js, tRPC, Prisma ORM, PostgreSQL, Multi-LLM Integration*
- Engineered an AI personalized recommendation engine synthesizing user interests, budget parameters, and historical patterns to generate curated choices with a 95% target recommendation accuracy.
- Architected a high-availability fallback abstraction layer supporting Gemini, Claude, and OpenAI that maintained 100% system uptime even during external API rate limits and network outages.
- Designed decoupled, asynchronous execution chains with tRPC, reducing frontend UI rendering bottlenecks by 40% and accelerating the processing of automated calendar occasion reminders.

Movie Assistant | *Next.js, React, Node.js, Express, PostgreSQL, Redis* — Live: vercel.app
- Developed a full-stack content personalization engine featuring a caching tier with Redis, reducing round-trip database queries and latency overheads by over 60%.
- Configured absolute JWT-based authentication schemes alongside modular Prisma relational schemas engineered to seamlessly index and retrieve up to 10k+ user watchlist records under 50ms.
- Optimized asynchronous controller operations to protect ingestion pathways, handling up to 500+ concurrent requests/sec without experiencing third-party API rate limit bottlenecks.

NoteCast - PDF to Podcast Converter | *Flask, React, Ollama LLM, Edge TTS, FFmpeg* — Code: Frontend | Backend
- Architected an end-to-end processing pipeline that extracts multi-page raw PDF content, leverages local LLMs for dialogue scripting, and synthesizes high-fidelity dual-voice audio outputs under 45 seconds.
- Leveraged Flask asynchronous worker architectures with thread pool optimization, resulting in a 4x reduction in overall CPU utilization during intense, parallel AI compilation workloads.
- Integrated system-level FFmpeg configurations to process real-time streaming audio stitching, decreasing audio file buffer allocation requirements and compilation overhead by 30%.

Technical Competitions

IndustriAI 24hr Hackathon - IIT Madras — 5-6 Jan 2024
- Proposed an ML-based conceptual solution utilizing XGBoost classification with SHAP-based interpretability to predict loan repayment risks based on structural financial profiles.
- Designed an adjustable parameter weighting model architecture designed to evaluate variable dependencies and mitigate bias during credit evaluations.

Research

Published paper: **"A Multi-Class Parasite Detection Framework in Blood Smears Using EfficientNet-B3 and Multi-Phase Fine-Tuning"** in Springer Nature (MAiTRI 2026). Developed a deep learning framework applying multi-phase fine-tuning, morphological feature extraction, and label smoothing to overcome class imbalance; achieved a **98.5% accuracy** and a **0.982 macro F1 score** in multi-class identification of blood-borne pathogens (malaria, babesiosis, leishmaniasis) and leukocytes.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, React, Next.js, Node.js, Express, Flask
Databases and Cloud Tools: PostgreSQL, SQL, Redis, Prisma ORM, Git, tRPC, Docker
Core Computer Science Foundations: Data Structures and Algorithms (DSA), System Design, RESTful APIs, Machine Learning